

Control work 1, and work on the mistakes

The basic geometric figures, tested — and the first chapter closed.

LESSON 13–14

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 1

S8 5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Use the additive properties of segments and angles under time.
- Classify angles and find complements and supplements.
- Use vertical and adjacent angles at a crossing.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 3–35
Cambridge	Stage 8, pp. 48–56
Workbook	Control paper M1

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	C lies between A and B ; $AC = 14$, $AB = 37$: find CB	4	L3–4
2	$AB = 26\text{ cm}$: find AM for the midpoint M	3	L3–4
3	Classify 35° , 90° , 142° , 200°	4	L5–6
4	Find the complement and supplement of 48°	4	L5–6
5	Two lines cross; one angle is 71° : find the other three	5	L9–10
6	Compute 0.42×0.5 and $3.6 \div 0.04$	5	L7–8, 11–12

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for writing $AB = AC + CB$ before substituting; Q3 one for the reflex angle; Q5 two for saying which pairs are vertical; Q6 two for the decimal places.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
subtraction reversed	$CB = 37 + 14$	$CB = 37 - 14$
unit missing	13	13 cm
200°	“obtuse”	reflex
complement and supplement swapped	132°	42° and 132° , named correctly
vertical taken as supplementary	109°	71°
decimal places miscounted	$0.42 \times 0.5 = 2.1$	0.21
only the divisor multiplied	$3.6 \div 4 = 0.9$	$360 \div 4 = 90$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter I as one map

BLOCK	THE SENTENCE
the basic figures	point, line and plane are undefined; everything else follows
the segment	$AB = AC + CB$ when C is between
the midpoint	$AM = MB = \frac{AB}{2}$
the angle	two rays from one vertex, measured in degrees
the five kinds	acute, right, obtuse, straight, reflex
at a crossing	adjacent add to 180° , vertical are equal
perpendicular	the shortest distance from a point to a line
decimals	count the places when multiplying; make the divisor whole when dividing

LOOKING FORWARD

The next chapter leaves geometry for a while and introduces **negative numbers** — the first real widening of the number system since fractions. The four operations all have to be learnt again, with signs.

02 Worked examples

EXAMPLE 1 Model answer, Q1: $AC = 14$, $AB = 37$.

① $AB = AC + CB$

The formula first.

② $37 = 14 + CB$

③ $CB = 23$

④ With the unit: 23 cm .

ANSWER

$$CB = 23\text{ cm}$$

EXAMPLE 2 Model answer, Q5: one angle at a crossing is 71° .

① Adjacent: $180^\circ - 71^\circ = 109^\circ$.

② Vertical to the first: 71° .
Equal.

③ The fourth: 109° .

④ Total 360° ✓

ANSWER

 $109^\circ, 71^\circ, 109^\circ$ **EXAMPLE 3** Model answer, Q6: 0.42×0.5 and $3.6 \div 0.04$.

① $42 \times 5 = 210$; places $2 + 1 = 3$.

② $= 0.210 = 0.21$

③ $3.6 \div 0.04 = 360 \div 4$
Both $\times 100$.

④ $= 90$

ANSWER

 0.21 and 90 **03** Interactive model

Return the papers with Q3 unmarked and let the class decide together whether 200° is obtuse or reflex; the vocabulary sticks when it is argued over.

Chapter I in twelve questions

Which is undefined?

segment

point

angle

triangle

Through two points there passes:

no line

one line

two lines

many

Which has two ends?

a line

a ray

a segment

a plane

$AB = AC + CB$ needs C :

anywhere

between A and B

outside

on a ray

A right angle is:

45°

90°

180°

360°

200° is:

obtuse

straight

reflex

acute

Complementary angles add to:

90°

180°

270°

360°

Vertical angles are:

supplementary

equal

complementary

unrelated

The distance to a line is measured:

any way

along the perpendicular

to the nearest point drawn

in degrees

0.3×0.4 equals:

1.2

0.12

0.012

12

$6 \div 0.1$ equals:

0.6

6

60

600

0.7 and 0.65:

0.65 larger

0.7 larger

equal

cannot tell

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Segment	Kesma	Отрезок
Angle	Burchak	Угол
Midpoint	O'rta nuqta	Середина
Bisector	Bissektrisa	Биссектриса
Adjacent angles	Qo'shni burchaklar	Смежные углы
Vertical angles	Vertikal burchaklar	Вертикальные углы
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 begins with:

the answer

the formula

a guess

a measurement

Q3 includes which unusual kind?

acute

right

reflex

straight

In Q5 the vertical angle is:

71°

109°

90°

289°

Q6 needs, for the product:

an estimate only

the count of decimal places

a calculator

rounding

For the division in Q6, multiply:

only the divisor

only the dividend

both by 100

neither

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $AC = 14, AB = 37$: CB

ANSWER 23

2. $AB = 26$: AM

ANSWER 13

3. Classify 35°

ANSWER Acute

4. Classify 142°

ANSWER Obtuse

5. Classify 200°

ANSWER Reflex

6. Complement of 48°

ANSWER 42°

7. Supplement of 48°

ANSWER 132°

Medium · the standard question

7 PROBLEMS

1. One angle 71° : the other three

ANSWER $109^\circ, 71^\circ, 109^\circ$

2. 0.42×0.5

ANSWER 0.21

3. $3.6 \div 0.04$

ANSWER 90

4. $AB = 40$ with $AC : CB = 3 : 5$: AC

ANSWER 15

5. Adjacent angles in the ratio $1 : 3$

ANSWER 45° and 135°

6. Bisector of 126°

ANSWER 63° each

7. Ascending: $0.5, 0.45, 0.505$

ANSWER $0.45, 0.5, 0.505$

Hard · stretch and reason

7 PROBLEMS

1. An angle is 5 times its complement

ANSWER 75°

2. $AB = 36$, M the midpoint of AB , N of MB : AN

ANSWER 27

3. Four angles at a point in the ratio $2 : 3 : 4 : 3$

ANSWER $60^\circ, 90^\circ, 120^\circ, 90^\circ$

4. 0.125×0.8

ANSWER 0.1

5. $2.4 \div 0.015$

ANSWER 160

6. A point 7 from a line and 24 along it from the foot

ANSWER 25

7. Two adjacent angles whose bisectors are 90° apart: always?

ANSWER Yes

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the next chapter begins.

1. C lies between A and B with $AC = 19$ and $AB = 45$. Find CB .
2. Find the complement and the supplement of 27° .
3. Two lines cross and one angle is 124° . Find the other three.
4. Compute 0.36×0.5 and $4.5 \div 0.09$.
5. Classify $89^\circ, 90^\circ, 91^\circ$ and 181° .